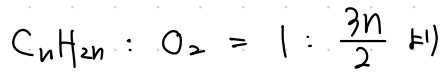
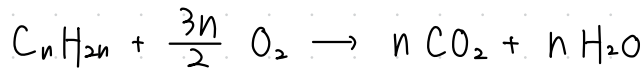
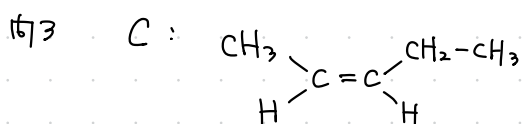
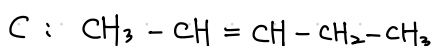
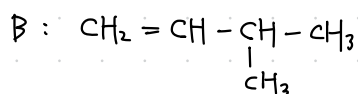
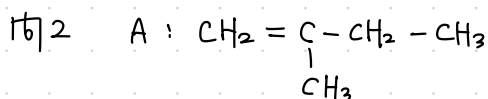
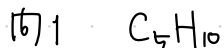
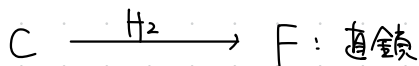
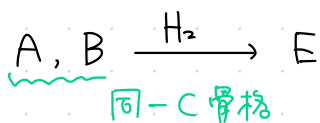
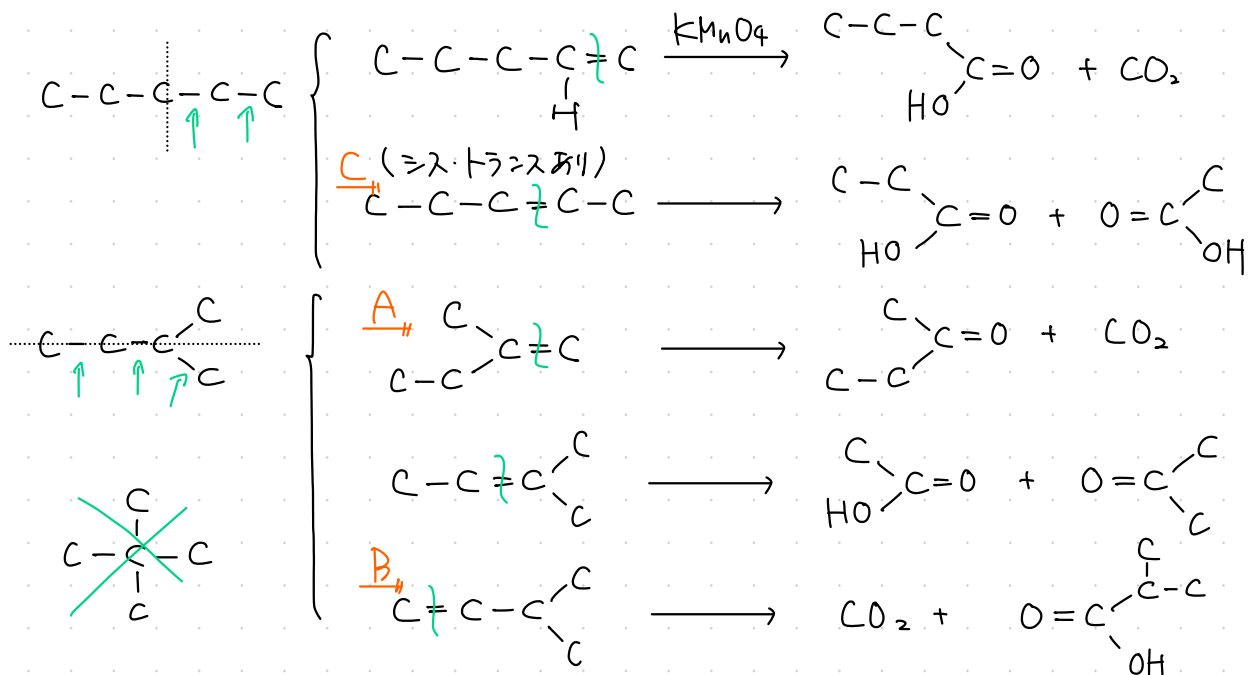
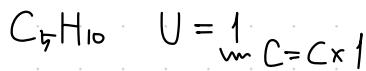


2-1

問1 (a), (b) は) A, B, C はアルケン C_nH_{2n}



$$1 \text{ mol} \times \frac{3n}{2} = \frac{168 \text{ L}}{22.4 \text{ L/mol}} \therefore n = 5 \quad \underline{C_5H_{10}}$$

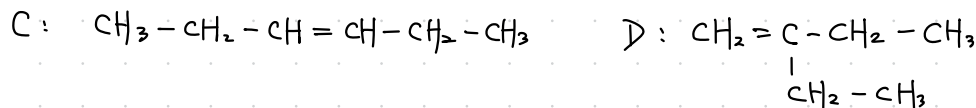
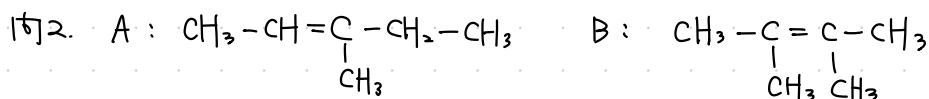


沸点 $\rightarrow$ 分子間力 $\left\{ \begin{array}{l} \text{水素結合} \\ \text{ファンデルワールス力} \end{array} \right.$

$\Rightarrow$ ス型の方が極性 $\odot \rightarrow$ 分子間力 $\odot$

2-2

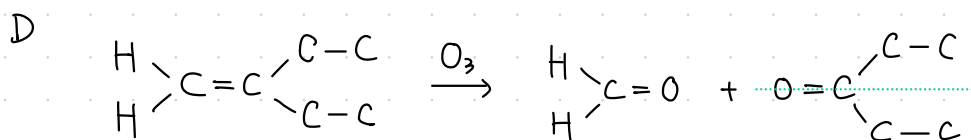
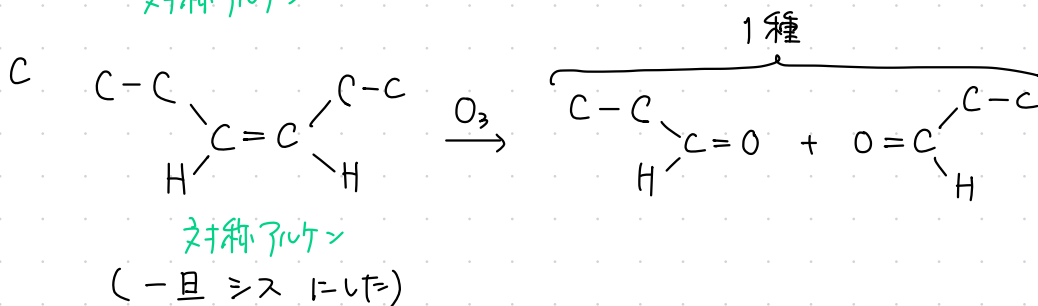
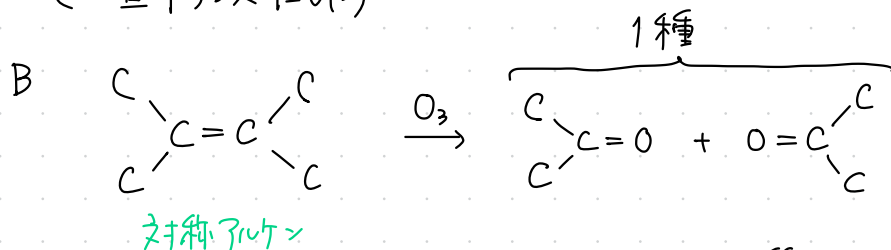
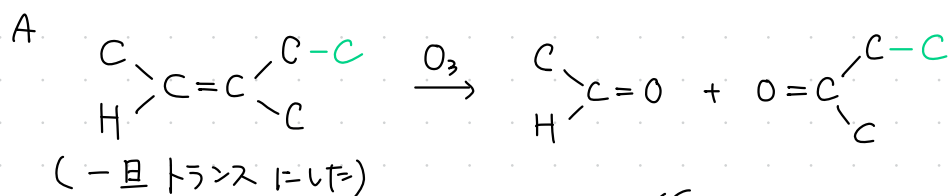
問1 C_6H_{12}



問3 $A \neq C$



$$\frac{2.1 \text{ g}}{14n \text{ g/mol}} = \frac{4.0 \text{ g}}{160 \text{ g/mol}} \quad \therefore n = 6 \quad \underline{C_6H_{12}}$$



2-3

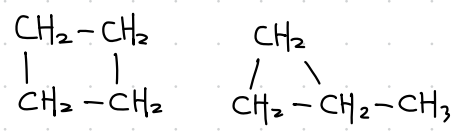
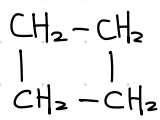
問1 A $\text{CH}_2=\text{CH}-\text{CH}_2-\text{CH}_3$ 1-ブテン

B $\text{CH}_3-\text{CH}=\text{CH}-\text{CH}_3$ 2-ブテン

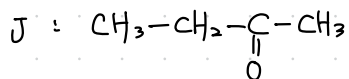
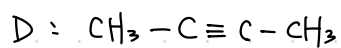
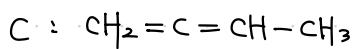
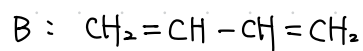
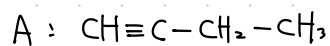
C $\text{CH}_3-\text{CH}_2-\text{CH}_2-\text{CH}_3$ ブタン

問2 D: 2 E: 3 (メソ体あり)

問3

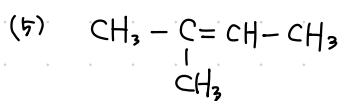
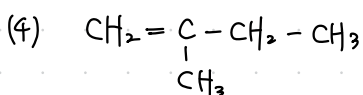
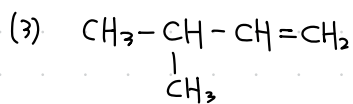


2 - 4



2 - 5

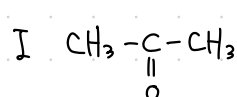
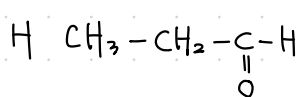
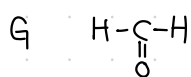
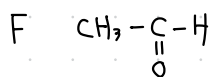
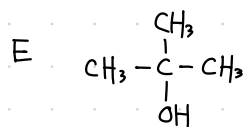
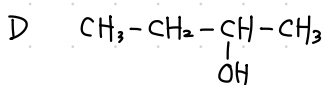
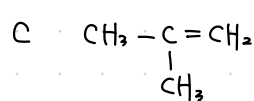
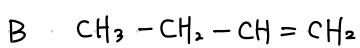
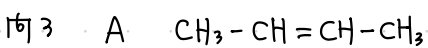
(1) 5 (2) 1



2 - 6

161 102

162 C_4H_8



164 F

2-7

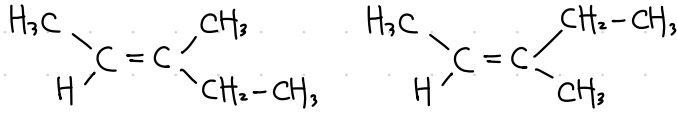
問1 ア 水素 イ 酸素 ウ ヒドロキシ基 エ ヨードホルム

問2 C C_4H_8O D C_2H_4O

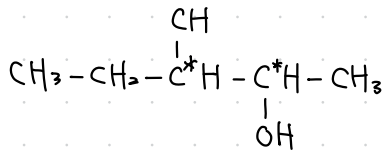
問3



問4



問5



問2.3

<C1=72.2>

$C_xH_yO_z$ と仮定,

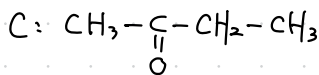
$x = \frac{72.2 \times \frac{66.7}{100}}{12 \text{ g/mol}} = 4$ ← $\frac{2}{3}$

$y = \frac{72.2 \times \frac{11.2}{100}}{1.0 \text{ g/mol}} = 8$ ← $\frac{1}{9}$

$z = \frac{72.2 \times \frac{22.1}{100}}{16 \text{ g/mol}} = 1$ ← $\frac{2}{9}$

$\therefore C: C_4H_8O_2$ (U=1)

$\boxed{\exists}+$, Redあり
7トシ

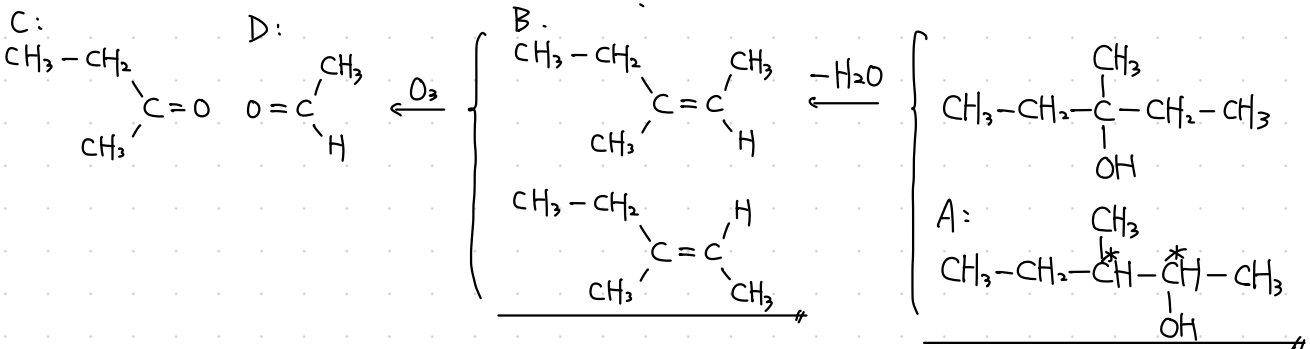
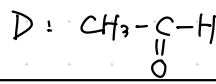


<D1=72.2>

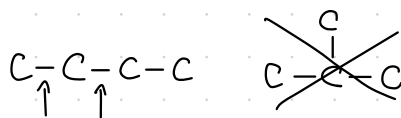
同様

D: C_2H_4O (U=1)

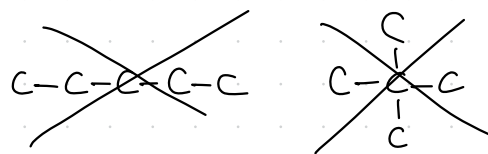
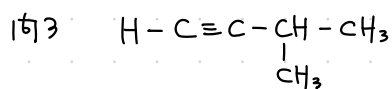
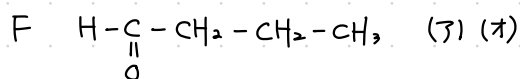
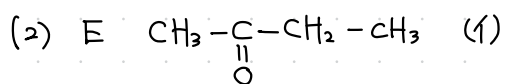
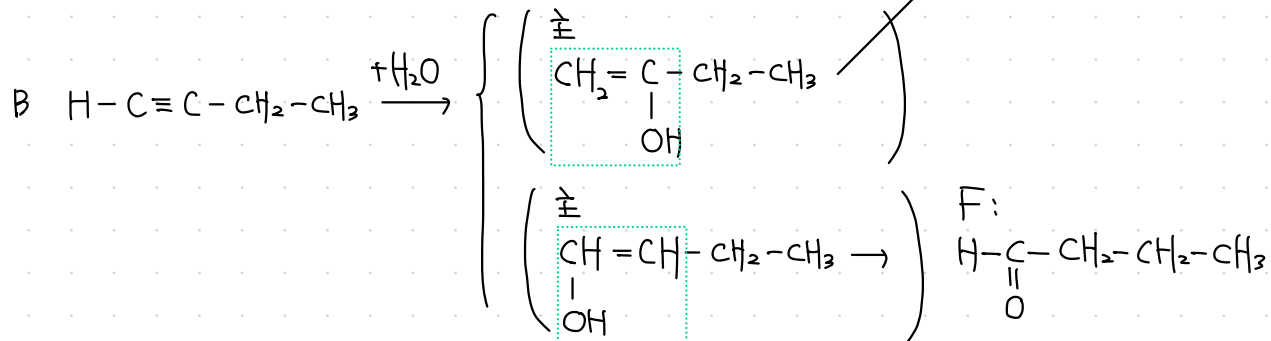
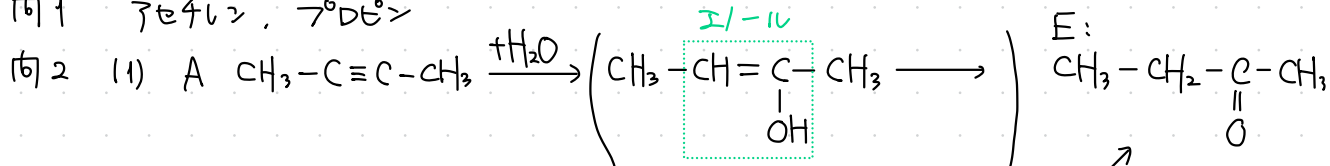
$\boxed{\exists}+$, Redあり
アセトアルデヒド



2-8



問1 了せし、7000



問4.

